

Lecture 5: Constrained Optimization II

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This lecture focuses on two other aspects of the Lagrangian approach:

- ▶ the sensitivity of the optimal value of the objective function to changes in the parameters
- ▶ the constraint qualifications that are a subtle but necessary hypothesis in the Lagrangian approach

- 1 The Meaning of the Multiplier
- 2 Envelope Theorems
- 3 Smooth Dependence on the Parameters
- 4 Constraint Qualifications

Consider the simplest problem with one equality constraint:

$$\max f(x, y) \quad (1)$$

$$\text{s.t. } h(x, y) = a \quad (2)$$

Theorem 19.1

- ▶ Let f and h be C^1 functions of two variables. For any fixed value of the parameter a , let $(x^*(a), y^*(a))$ be the solution of problem (1)–(2) with corresponding multiplier $\mu^*(a)$.
- ▶ Suppose that x^* , y^* and μ^* are C^1 functions of a and that NDCQ holds at $(x^*(a), y^*(a))$. Then

$$\mu^*(a) = \frac{d}{da} f(x^*(a), y^*(a))$$

Example 19.1: Previously, we found that a maximizer of $f(x_1, x_2) = x_1^2 x_2$ on the constraint set $2x_1^2 + x_2^2 = 3$ is $x_1 = x_2 = 1$, with multiplier $\mu = 0.5$.

- ▶ The maximum value of f is $f^* = f(1, 1) = 1$
- ▶ Redo the problem, this time using the constraint

$$2x_1^2 + x_2^2 = 3.3$$

- ▶ The same computation yields the solution $x_1 = x_2 = \sqrt{1.1}$, with maximum value $f^* = (1.1)^{3/2} \approx 1.1537$, an increase of 0.1537 over the original f^*

Example 19.1:

- ▶ On the other hand, Theorem 19.1 predicts that changing the right-hand side of the constraint by 0.3 unit would change the maximum value of the objective function by roughly

$$0.3\mu = 0.3 * 0.5 = 0.15 \text{ unit,}$$

an approximation correct to two decimal places

Theorem 19.2

Let $f, h_1, \dots, h_m$ be C^1 functions on $\mathbb{R}^n$. Let $\mathbf{a} = (a_1, \dots, a_m)$ be an m -tuple of exogenous parameters, and consider the problem of maximizing $f(x_1, \dots, x_n)$ subject to

$$h_1(x_1, \dots, x_n) = a_1, \dots, h_m(x_1, \dots, x_n) = a_m.$$

Let $x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a})$ denote the solution of the problem with corresponding Lagrange multipliers $\mu_1^*(\mathbf{a}), \dots, \mu_m^*(\mathbf{a})$. Suppose further that the x_i^* 's and μ_j^* 's are differentiable functions of $(a_1, \dots, a_m)$ and that NDCQ holds. Then, for each $j = 1, \dots, m$,

$$\mu_j^*(a_1, \dots, a_m) = \frac{\partial}{\partial a_j} f(x_1^*(a_1, \dots, a_m), \dots, x_n^*(a_1, \dots, a_m))$$

Theorem 19.3

Let $\mathbf{a}^* = (a_1^*, \dots, a_k^*)$ be a k -tuple. Consider the problem of maximizing $f(x_1, \dots, x_n)$ subject to

$$g_1(x_1, \dots, x_n) \leq a_1^*, \dots, g_k(x_1, \dots, x_n) \leq a_k^*.$$

Let $x_1^*(\mathbf{a}^*), \dots, x_n^*(\mathbf{a}^*)$ denote the solution of the problem with corresponding Lagrange multipliers $\lambda_1^*(\mathbf{a}^*), \dots, \lambda_k^*(\mathbf{a}^*)$. Suppose that as $\mathbf{a}$ varies near $\mathbf{a}^*$, $x_1^*, \dots, x_n^*$ and $\lambda_1^*, \dots, \lambda_k^*$ are differentiable functions of $(a_1, \dots, a_k)$ and that NDCQ holds at $\mathbf{a}^*$. Then, for each $j = 1, \dots, k$,

$$\lambda_j^*(a_1^*, \dots, a_k^*) = \frac{\partial}{\partial a_j} f(x_1^*(a_1^*, \dots, a_k^*), \dots, x_n^*(a_1^*, \dots, a_k^*))$$

Think of the objective function as the profit function of a firm and think of a_j as the amount of available input in the production process. Then $\lambda_j = \partial f / \partial a_j$ is called the *internal value*, or *shadow price* of input j :

- ▶ It tells how valuable another unit of certain input would be to the firm's profits
- ▶ It tells the maximum amount the firm would be willing to pay to acquire another unit of the input

Example 19.2 Previously, we computed that the maximizer of xyz on the constraint set

$$x + y + z \leq 1, x \geq 0, y \geq 0, z \geq 0$$

is $x = y = z = 1/3$, where $xyz = 1/27$. The four multipliers are $1/9, 0, 0$ and 0 respectively.

- (a) If we change the first constraint to $x + y + z \leq 0.9$, we compute that the solution occurs at $x = y = z = 0.3$, where $xyz = 0.027$. Theorem 19.3 predicts that the new optimal value would be $\frac{1}{27} + \frac{1}{9} \cdot \left(-\frac{1}{10}\right) \approx 0.0259$, an estimate that is off by only 0.0011 or 4%.
- (b) If, instead, we change the second constraint from $x \geq 0$ to $x \geq 0.1$, we do not change the solution or the optimum value because the new region is a subset of the old region and it still contains the optimal point for the old region. This result is consistent with Theorem 19.3 since the multiplier for the (nonbinding) constraint $x \geq 0$ was zero.

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- ▶ Previous theorems are special cases of a class of theorems which describe how the optimal value of the objective function in a parameterized optimization problem changes as one of the parameter changes.
- ▶ Such theorems are called **Envelope theorems**.

Theorem 19.4

Let $f(\mathbf{x}; a)$ be a C^1 function of $\mathbf{x} \in \mathbb{R}^n$ and the scalar a . For each choice of the parameter a , consider the unconstrained maximization problem: $\max_{\mathbf{x}} f(\mathbf{x}; a)$. Let $\mathbf{x}^*(a)$ be a solution of this problem. Suppose that $\mathbf{x}^*(a)$ is a C^1 function of a . Then

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a)$$

Example 19.3 Consider the problem

$$\max_x f(x, a) = -a^3x^4 + 15x^3 - e^ax^2 + 17$$

around $a = 1$. Since f is a quartic polynomial in x with a negative leading coefficient, when $a = 1$, $f \rightarrow -\infty$ as $x \rightarrow \pm\infty$. So f admits a finite global maximizer $x^*(a)$ for each value of a near 1. Hence,

$$\frac{d}{da}f(x^*(a), a) = \frac{\partial}{\partial a}f(x^*(a), a) = -3a^2x^{*4} - e^ax^{*2} < 0$$

Without solving for the optimal $x^*(a)$, we can tell that as a increases beyond 1, $f(x^*(a), a)$ is a decreasing function of a . The peak of the graph of the function decreases as a increases.

Example 19.4 What will be the effect of a unit increase in a on the value of

$$\max_x f(x, a) = -x^2 + 2ax + 4a^2$$

for each a ? Since $f'(x) = -2x + 2a = 0$, $x^*(a) = a$. Then

$$f(x^*(a), a) = f(a, a) = -a^2 + 2a * a + 4a^2 = 5a^2,$$

which will increase at a rate of $10a$ as a increases. Instead, if we apply the envelope theorem,

$$\frac{df^*}{da} = \frac{\partial f}{\partial a}(x^*(a), a) = 2x + 8a = 10a$$

since $x^*(a) = a$.

Example 19.5

- ▶ A silicon Valley firm produces an output of microchips denoted by y and has a cost function $c(y)$ with $c'(y) > 0$ and $c''(y) > 0$.
- ▶ Of the chips it produces, a fraction $1 - \alpha$ are unavoidably defective and cannot be sold.
- ▶ Working chips can be sold at price p , and the microchip market is highly competitive.

How will an increase in production quality affect the firm's profit?

Example 19.5 The firm's profit function is given by

$$\pi(p, \alpha) = \max_y [p\alpha y - c(y)]$$

- ▶ The conditions on the cost function guarantee that there is a nonzero profit-maximizing output $y^*(\alpha)$ which depends smoothly on α .
- ▶ The derivative of optimal profit π with respect to α is

$$\frac{d\pi}{d\alpha} = \frac{\partial}{\partial \alpha}(p\alpha y - c(y)) = py > 0$$

Theorem 19.5

Let $f, h_1, \dots, h_k: \mathbb{R}^n \times \mathbb{R}^1 \rightarrow \mathbb{R}^1$ be C^1 functions. Let $\mathbf{x}^*(a) = (x_1^*(a), \dots, x_n^*(a))$ denote the solution of the problem of maximizing $\mathbf{x} \rightarrow f(\mathbf{x}, a)$ on the constraint set

$$h_1(\mathbf{x}; a) = 0, \dots, h_k(\mathbf{x}; a) = 0$$

for any fixed choice of the parameter a . Suppose that $\mathbf{x}^*(a)$ and the Lagrangian multipliers $\mu_1(a), \dots, \mu_k(a)$ are C^1 functions of a and that the NDCQ holds. Then,

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial L}{\partial a}(\mathbf{x}^*(a), \mu(a); a).$$

Example 19.6 Consider the problem

$$\begin{aligned} \max \quad & f(x, y) = xy \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \end{aligned}$$

Now change the constraint to $x^2 + 1.1y^2 \leq 1$. If we write the constraint as $x^2 + ay^2 \leq 1$, the Lagrangian for the parameterized problem is $L(x, y, \lambda; a) = xy - \lambda(x^2 + ay^2 - 1)$.

- ▶ The solution for the original ($a = 1$) problem was $x = y = \frac{1}{\sqrt{2}}$, $\lambda = \frac{1}{2}$
- ▶ The Envelope Theorem tells us that as a changes from 1 to 1.1, the optimal value of f changes by approximately $\frac{\partial L}{\partial a} \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}; 1 \right) * 0.1$
- ▶ Since $\frac{\partial L}{\partial a} = -\lambda y^2 = -1/4$, the optimal value will decrease by approximately $0.1 * (1/4) = 0.025$ to 0.475
- ▶ Calculate directly that the solution to the new problem is $x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2.2}}$, with $f^* \approx 0.4767$.

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Theorem 19.9

Consider the problem of maximizing $f(x; a)$ subject to $h_1(x; a) = 0$, $h_2(x; a) = 0$, $\dots$, $h_k(x; a) = 0$. Let $x^*(a)$ be the solution of the parameterized constrained maximization problem and let $\mu^*(a)$ be the corresponding Lagrange multiplier. If the Hessian matrix of the Lagrangian is nonsingular at the point $(x^*(a_0), \mu^*(a_0); a_0)$, then:

- 1 $x^*(a)$ and $\mu^*(a)$ are C^1 functions of a at $a = a_0$; and
- 2 the NDCQ holds at $(x^*(a_0), \mu^*(a_0); a_0)$.

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- ▶ In Lecture 4, we talked about NDCQ.
- ▶ There are theorems that do not impose constraint qualifications.

Theorem 19.10

Let f and h be C^1 functions of two variables. Suppose that $\mathbf{x}^* = (x_1^*, x_2^*)$ is a solution of the problem of maximizing $f(x_1, x_2)$ subject to $\{(x_1, x_2) : h(x_1, x_2) = c\}$.

Construct the Lagrangian $L(x_1, x_2, \mu_0, \mu_1) = \mu_0 f(x_1, x_2) - \mu_1 [h(x_1, x_2) - c]$, including a multiplier μ_0 for the objective function. Then, there exist multipliers μ_0^* and μ_1^* such that:

- (a) μ_0^* and μ_1^* are not both zero,
- (b) μ_0^* is either 0 or 1, and
- (c) the quadruple $(x_1^*, x_2^*, \mu_0^*, \mu_1^*)$ satisfies the equations

$$\frac{\partial L}{\partial x_1} = \mu_0^* \frac{\partial f}{\partial x_1}(x_1^*, x_2^*) - \mu_1^* \frac{\partial h}{\partial x_1}(x_1^*, x_2^*) = 0,$$

$$\frac{\partial L}{\partial x_2} = \mu_0^* \frac{\partial f}{\partial x_2}(x_1^*, x_2^*) - \mu_1^* \frac{\partial h}{\partial x_2}(x_1^*, x_2^*) = 0,$$

$$\frac{\partial L}{\partial \mu_1} = c - h(x_1^*, x_2^*) = 0.$$

Theorem 19.11

Suppose that $f, g_1, \dots, g_k$ are C^1 functions of n variables. Suppose that $\mathbf{x}^*$ is a local maximizer of f on the constraint set defined by the k inequalities $g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k$. Form the Lagrangian

$$L(x_1, \dots, x_n, \lambda_0, \lambda_1, \dots, \lambda_k) = \lambda_0 f(\mathbf{x}) - \lambda_1 [g_1(\mathbf{x}) - b_1] - \dots - \lambda_k [g_k(\mathbf{x}) - b_k],$$

with a multiplier λ_0 for the objective function. Then, there exist multipliers $\lambda^* = (\lambda_0^*, \lambda_1^*, \dots, \lambda_k^*)$ such that:

- (a) $\frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0,$
- (b) $\lambda_1^* [g_1(\mathbf{x}^*) - b_1] = 0, \dots, \lambda_k^* [g_k(\mathbf{x}^*) - b_k] = 0,$
- (c) $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0,$
- (d) $g_1(\mathbf{x}^*) \leq b_1, \dots, g_k(\mathbf{x}^*) \leq b_k,$
- (e) $\lambda_0^* = 0$ or 1 , and
- (f) $(\lambda_0^*, \lambda_1^*, \dots, \lambda_k^*) \neq (0, 0, \dots, 0).$